A. MAXIMUM PAYLOAD WHILE MAINTAINING STABLE FLIGHT.

For the given purpose the candidate aerofoils are:

1) CAMBERED AIRFOIL:

A fixed curved profile that produces lift efficiently over its intended operating range. It is comparatively simple and is a useful baseline for a payload-carrying aircraft.

2) HIGH-LIFT AIRFOIL:

A wing with flaps and possibly leading-edge slats that increase lift when needed, particularly during take-off and landing. The increased lift generally comes with extra drag and mechanical complexity.

3) EFFICIENT CAMBERED AIRFOIL- VARIABLE/MORPHING CAMBERED AIRFOIL:

The wing changes its curvature to suit different flight conditions. It can use a more efficient shape in cruise and a higher-lift shape at low speeds, if properly designed and tested.

COMPARING THE CANDIDATE AIRFOILS ON XFLR5 AND XFOIL WITH RELEVANT REYNOLD NUMBER:

WE KEEP THE REYNOLD NUMBER AND ALL OTHER PARAMETERS SAME FOR ALL THREE CANDIDATES TO MAKE COMPARISON MORE FAIR.

The candidate air foils in the XFLR5:

1] **For Conventional Cambered Airfoil we use NACA2412**:

Conventional cambered baseline. The four-digit designation indicates 2% maximum camber, located at 40% chord, and 12% maximum thickness.

2] **For High-lift Airfoil we use NACA4415**:

A more highly cambered baseline for examining lift and pitching-moment trade-offs. This is not a complete multi-element high-lift system; it is a single aerofoil with greater camber and thickness.

3] **For Efficient Cambered Airfoil with laminar flow candidate we use NACA64-415**:

A representative six-series laminar-flow candidate. Its performance depends on the actual Reynolds number, surface condition and operating lift coefficient.

REYNOLDS NUMBER:

This is essential because an aerofoil can behave differently at different speeds and sizes.

$$Re = \frac{\rho V c}{\mu}$$

Where ,

ρ = air density in kg/m³

V = airspeed in m/s

c = chord length in metres

μ = dynamic viscosity in Pa*s

By keeping in mind the above mentioned condition I have taken these as some suitable values of the parameters to calculate Reynolds number:

Flight Speed = 11.68 m/s

Chord = the distance of the leading edge from the trailing edge = 0.25 m

here are some values which actually depend on the air conditions,

Air Density = 1.225 kg/m³

Dynamic Viscosity = 1.789 * 10^(-5) Pa * s

CALCULATION

$$Re = \frac{1.225 \times 11.68 \times 0.25}{1.789 \times 10^{-5}}$$

FOR RUNNING 2D STIMULATION OF AIRFOIL THE VALUES TAKEN IN ANALYSIS SETTINGS ARE:

XFLR5 setting	Value to enter	Why we choose it
Analysis type	Type 1 — α (angle of attack)	Measures how lift, drag and pitching moment change as the airfoil's angle changes.
Reynolds number (Re)	200,000	A useful starting point for comparing airfoils on a small, low-speed model. The correct value depends on your aircraft's speed and wing chord.
Mach number	0.0	Suitable for an initial low-speed comparison where compressibility is negligible.
Ncrit	9	A commonly used baseline for airfoil analysis; keep it identical for all three profiles.
Minimum angle of attack	−5°	Shows performance at negative and low angles.
Maximum angle of attack	15°	Helps investigate higher-lift conditions, although the solver may not converge near stall.
Angle increment	0.5°	Gives enough detail to compare the curves without making the first test unnecessarily complicated.
Store operating points	Checked	Saves the individual calculated conditions for later inspection.
Forced transition	Leave at the same default for all three	Keeps the comparison consistent; changing transition settings can change predicted drag.

We have compared the following ratios for the three candidate airfoils through the graphs formed with the help of XFLR5 and XFOIL:

- **C_L vs. angle of attack**: how lift changes as the airfoil tilts into the airflow.
- **C_D vs. angle of attack:** how much drag is predicted.

- **C_L / C_D vs. angle of attack:** aerodynamic efficiency.
- **C_M vs. angle of attack:** how pitching moment changes, which is relevant to the aircraft's trim and stability.

I have also included the trailing-edge flap withs values for moderate flap and the reasoning behind these values:

- **Hinge at 70% chord**
  This leaves the front 70% of the airfoil unchanged and makes the rear 30% the flap. It is a reasonable baseline for testing how additional camber changes lift and drag. The best hinge position still depends on the actual airfoil and flap geometry.

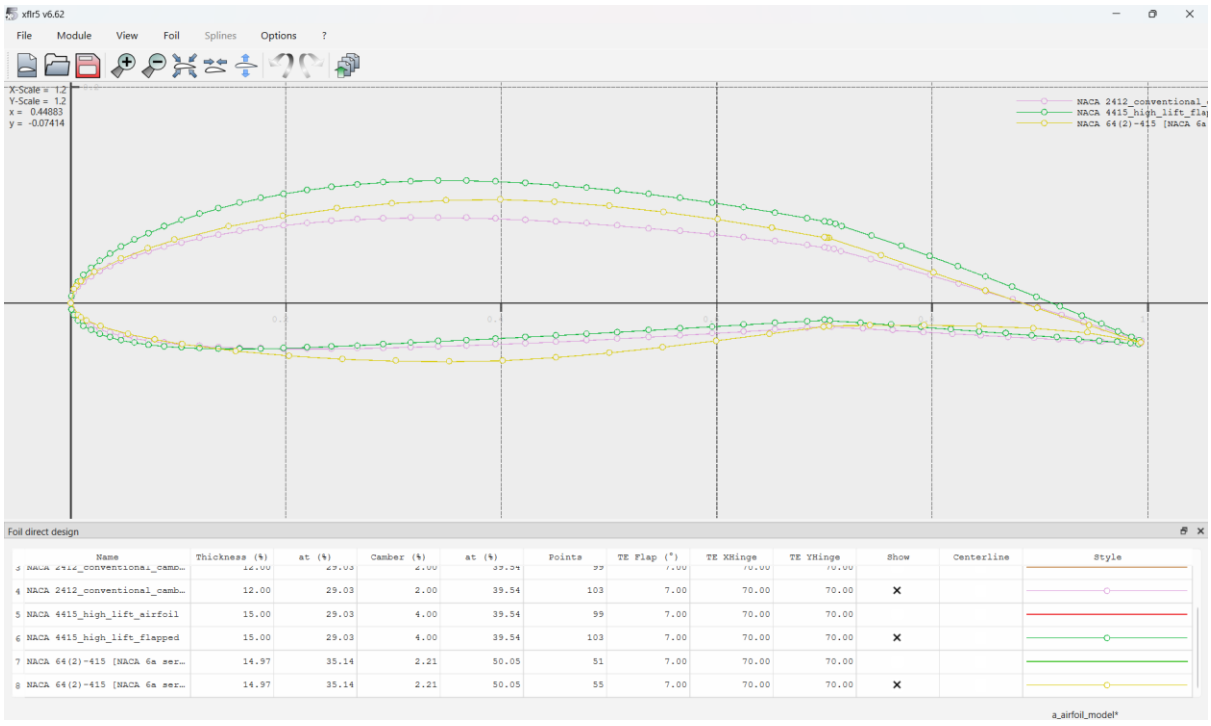
- **Deflection of 7° downward**
  A downward flap increases effective camber, which generally increases lift at a given angle of attack. A moderate initial deflection lets you measure that benefit while avoiding the larger drag increases that can occur at higher deflections.

- **Why not use a larger angle immediately?**
  More flap deflection does not automatically mean a better aircraft. It can increase drag, change pitching moment, and reduce the angle-of-attack margin before stall. Stable flight must be checked for the complete aircraft, not inferred from lift alone

THE FINAL COMPARISON: Let us first see the models of the three candidate airfoils given by XFLR5 given is the link of those models in XFLR5 in its default form .xfl:

a_airfoil_model.xfl

- Let us first see the models of the three candidate airfoils with flap:

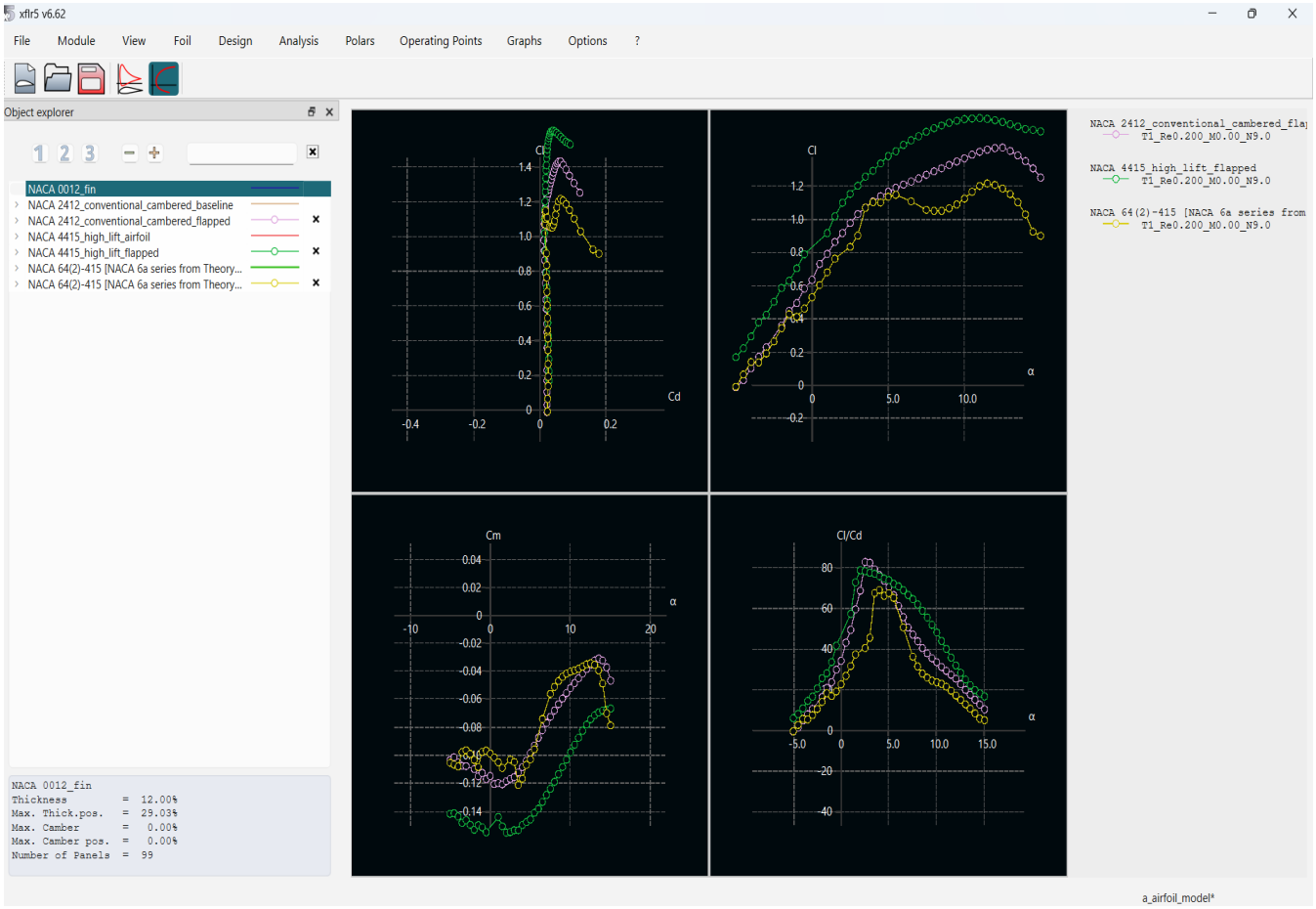


- Also look towards there the 4 important graphs related to them:

THE LINK OF THE 4 GRAPHS (.CSV FLIES) WITH FLAPPED AS WELL AS WITHOUT FLAPPED

graph_cl_cd.csv graph_cl_alpha.csv graph_cm_alpha.csv graph_clcd_alpha.csv

AIROFOIL



The final result of the compared values:

Table Flapped Airfoil Comparison

	Parameter	NACA 2412 flapped	NACA 4415 flapped	NACA 64(2)-415 flapped
A	Maximum thickness	12%	15%	14.97%
B	Maximum camber	2%	4%	2.21%
C	Trailing-edge flap deflection	7°	7°	7°
D	Hinge location	70% chord	70% chord	70% chord
E	Relative lift potential	Moderate	High	Moderate–high
F	Relative drag tendency	Moderate	Moderate–high	Low–moderate before separation
G	Main advantage	Balanced performance	Higher camber and lift potential	Potentially efficient cruise
H	Main limitation	Lower lift potential than the more cambered profile	May incur more drag	Laminar-flow benefits depend on conditions

Table How to Report the Four Graph Outputs

	Graph	What to record	What it tells you
1	C_L vs angle of attack	Lift coefficient at each angle	Lift generation and stall trend
2	C_D vs angle of attack	Drag coefficient at each angle	Aerodynamic resistance
3	C_L/C_D vs angle of attack	Lift-to-drag ratio	Aerodynamic efficiency
4	C_m vs angle of attack	Pitching-moment coefficient	Pitching tendency and stability considerations

THE FINAL CONCLUSION FROM THE ABOVE COMPARISON IS:

My final preliminary choice is **HIGH-LIFT CAMBERED AIRFOIL – NACA4415**. The main reasons are:

- **Greater lift potential**

  NACA 4415 has 4% maximum camber, compared with 2% for NACA 2412. Its greater camber can help generate more lift at a given angle of attack under suitable conditions. This can be useful when carrying additional payload.

- **More thickness for structural design**

  Its 15% thickness gives the wing designer more internal space for spars and structure than a 12%-thick profile, all else being equal. Actual strength depends on the spar design, materials, and wing construction.

- **Potentially useful at low flight speeds**

  Higher lift at a given speed can help a model support its weight at a lower speed or reduce the angle of attack required. The actual stall speed must be determined from the maximum lift coefficient and the aircraft's total weight and wing area.

NOTE: THERE EXIST ALSO MANY MORE FACTORS WHICH CAN AFFECT THE CAPACITY OF PAYLOAD AND STABILITY LIKE THE AIRCRAFTS WEIGHT, LONGITUDINAL STABILITY, CENTER OF GRAVITY, WING-TAIL ARRANGMENT AND THE ACTUAL FLIGHT CONDITIONS.

B. OPTIMIZED FOR MANEUVERABILITY (TURNS, ROLLS, LOOPS).

Maneuverability is an aircraft's ability to change its direction and orientation quickly, accurately, and controllably while remaining stable.

- Turns: Change the aircraft's flight direction.
- Rolls: Rotate the aircraft around its longitudinal axis.
- Loops: Perform a vertical circular flight path by changing pitch and managing lift and speed.

For this purpose the candidate airfoils are:

❖ **NACA 2412 — balanced baseline**

- Recommended baseline
- Moderate camber and 12% thickness.
- A useful starting point for predictable handling and general-purpose flying.
- Compare its lift curve, drag, and stall behaviour before choosing it.

❖ **NACA 23012 — alternative for balanced lift**

- 12% thickness, with a different camber distribution from NACA 2412.
- Worth testing for lift and pitching-moment characteristics.
- Its actual handling advantages depend on Reynolds number and the complete aircraft design.

❖ **NACA 0012 — symmetrical aerobatic candidate**

- Zero camber and 12% thickness.
- Similar basic geometry when the aircraft flies upright or inverted, at equal-magnitude positive and negative angles of attack in ideal conditions.
- A useful candidate for evaluating loops and inverted flight.

NOTE: HERE IT IS NOT THAT SIGNIFICANT TO ADD FLAP IN THE AIRFOIL AS OUR AIM IS TO OPTIMIZE MANEUVERABILITY WHICH IS NOT HEAVILY DEPENDED ON THE FACTOR OF HIGH LIFT COEFFICIENT.

For the Reynolds number the value remains same here also that is Re= 200000 in XFLR5.

The reasons are:

- **Suitable for a small RC aircraft**
- **Fair comparison of your airfoils**
- **Useful for evaluating maneuverability**

The suitable values for the 2d stimulation by XFLR5 and XFOIL:

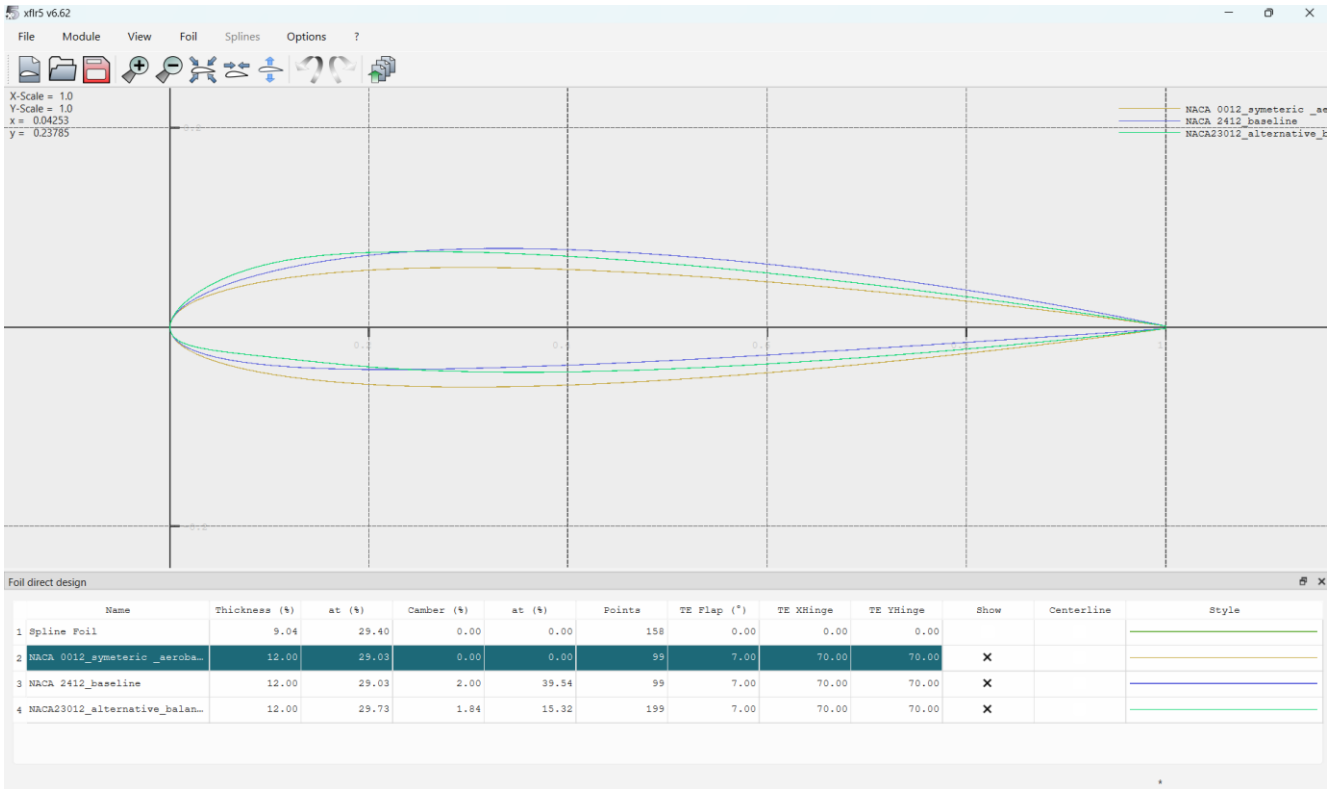
Parameter	Suggested starting setting	Why is this value chosen?
Reynolds number	**200,000**	A reasonable starting point for comparing airfoils on a small RC aircraft. It helps evaluate lift, drag, and stall behavior under the same conditions.
Mach number	**0.03–0.06**, depending on speed	Represents low-speed flight, where compressibility effects are generally small. The exact value should match the aircraft's flight speed.
Angle of attack	**−5° to +15°**, if the solver converges	Covers negative, moderate, and higher angles of attack, helping compare lift performance and identify the onset of stall.
Angle step	**0.5°**	Gives a reasonably detailed curve of lift, drag, and pitching moment without making the analysis excessively fine.
Airfoils	**NACA 2412, NACA 23012, NACA 0012**	NACA 2412 is a balanced cambered baseline; NACA 23012 provides an alternative camber distribution; NACA 0012 is symmetric and useful for investigating inverted-flight and aerobatic characteristics.

THE FINAL COMPARISON

Let us see the final comparison between the three candidate airfoils:

b_airfoil.xfl

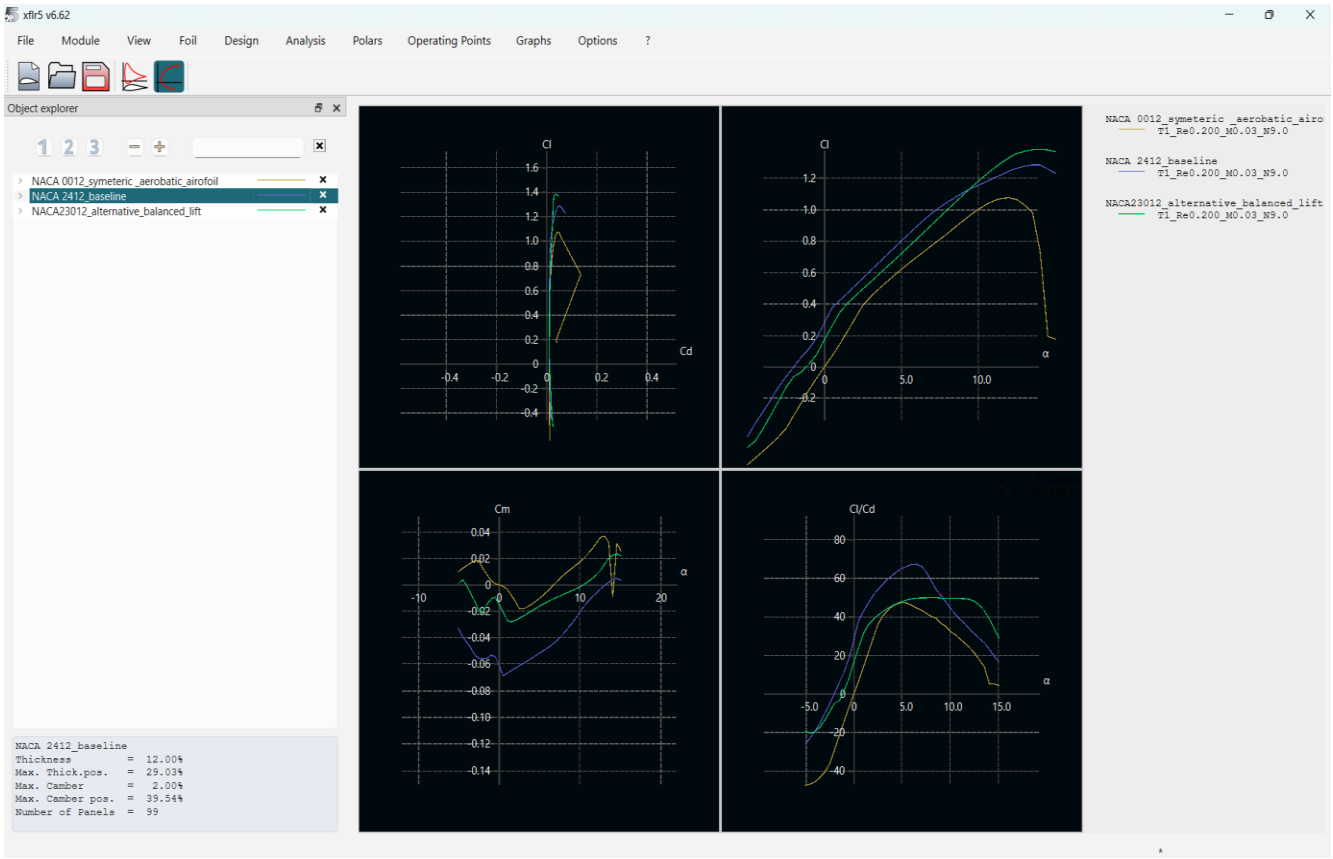
- Comparison between the models of the candidate airfoils:



- Also look towards there the 4 important graphs related to them:

THE LINK OF THE 4 GRAPHS (.CSV FLIES) WITHOUT FLAP AIRFOILS ARE GIVEN BELOW:

graph_b_clcd_alpha.csv graph_b_cm_alpha.csv grph_b_cl_alpha.csv grph_b_cl_cd.csv



The final result of the compared values:

Parameter	NACA 0012 (yellow)	NACA 2412 (blue)	NACA 23012 (green)
Maximum lift coefficient, $C_{L,\max}$	About 1.1	About 1.3	About 1.4
Angle near maximum lift	About 12°	About 12°	About 12°
Peak lift-to-drag ratio, C_L/C_D	About 48	About 67	About 50
Angle near peak C_L/C_D	About 4°	About 6°	About 7°
Drag polar, C_D vs C_L	Shows increasing drag as lift increases	Shows a favorable lift-to-drag trend	Shows high lift, with drag increasing at higher lift
Pitching moment, C_m	Varies with angle; appears less negative over parts of the range	More negative over much of the plotted range	Moderately negative over parts of the range
Main observed advantage	Symmetric profile; useful for studying inverted flight	Highest apparent peak aerodynamic efficiency	Highest apparent maximum lift
Main consideration	Lowest apparent maximum lift of the three	Good balance of lift and efficiency	Higher lift does not necessarily mean better efficiency

THE FINAL CONCLUSION:

My final preliminary choice is **NACA 2412**. The main reasons are:

- ❖ It offers a useful balance of lift and aerodynamic efficiency, and your XFLR5 graphs show the highest apparent peak C_L / C_D among the three candidates.

- ❖ The high C_L / C_D implies that how efficiently an airfoil produces lift relative to drag.

NOTE: THE PROPERTY OF MANEUVERABILITY ALSO DEPENDS ON FACTORS LIKE AILERONS, ELEVATOR AUTHORITY, WING LOADING AND COMPLETE AIRCRAFT DESIGN.

HOW AIRFOIL GEOMETRY AFFECTS THE AIRCRAFTS PERFORMANCE:

An airfoil is the cross-sectional shape of a wing. Its geometry determines how air flows around the wing and strongly influences the lift, drag, stall behavior, and pitching moment produced by the wing.

For an **RC plane**, airfoil geometry affects how much payload the aircraft can carry, how slowly it can fly, how efficiently it uses battery power, and how effectively it performs maneuvers such as turns, rolls, and loops.

The main geometric features are:

- ❖ **Camber**

Camber is the curvature of the airfoil's mean line.

- More positive camber generally produces more lift at a given angle of attack.
- It can help an RC plane generate lift at lower speeds.
- Excessive camber can increase drag and pitching moment.

Effect: Useful for payload carrying and low-speed flight, but the optimum depends on the operating conditions.

- ❖ **Thickness**

Thickness is the maximum distance between the upper and lower surfaces, expressed as a percentage of chord.

- A thicker airfoil can provide more room for structural strength and internal components.
- It can help accommodate a strong wing structure.
- Thickness also affects drag, flow separation, and pitching characteristics.

Effect: Useful when balancing structural strength and aerodynamic performance.

❖ Symmetric versus cambered shape

- A **cambered airfoil**, such as **NACA 2412**, typically produces positive lift at zero angle of attack.
- A **symmetric airfoil**, such as **NACA 0012**, produces approximately zero lift at zero angle of attack in ideal conditions.

Effect: Cambered airfoils are often useful for efficient normal flight, while symmetric airfoils are useful when similar aerodynamic behavior at positive and negative angles of attack is desirable, such as during inverted flight.

❖ Leading-edge and trailing-edge shape

The leading edge affects how smoothly the airflow accelerates around the front of the airfoil. The trailing edge influences how the flow leaves the airfoil.

Effect: These shapes influence drag, stall behavior, and the response to angle of attack.

Physics Behind The Airfoil Performance:

As an RC plane moves forward, air flows around its wings. The wing changes the airflow and creates a pressure distribution; the resulting aerodynamic forces include lift and drag.

LIFT IS EXPRESSED AS:

$$\boldsymbol{LIFT = L = \frac{1}{2}\rho V^2 S C_L}$$

DRAG IS EXPRESSED AS:

$$\boldsymbol{DRAG = D = \frac{1}{2}\rho V^2 S C_D}$$

Where:

- **L** = lift force
- **D** = drag force
- $\boldsymbol{\rho}$ = air density
- **V**= aircraft speed
- **S**= wing reference area
- $\mathbf{C_L}$ = lift coefficient
- $\mathbf{C_D}$ = drag coefficient

- A higher lift coefficient can help an RC plane support more weight at a given speed and wing area. However, if the airfoil generates too much drag, the motor must provide more power. If the angle of attack becomes too large, airflow separation can cause a stall and a loss of lift.
- In terms of stability; Airfoil geometry contributes to pitching moment, but overall aircraft stability also depends on the center of gravity, horizontal tail, tail volume, and wing configuration.

AIRFOILS GEOMETRY EFFECT ON THE CASE (A) AIRCRAFT:

Geometric feature	Preferred characteristic	Reason
Camber	Moderate to relatively high	Helps generate lift at lower speeds, depending on the airfoil and operating conditions.
Thickness	Sufficient for a strong wing structure	Helps accommodate the spar and structure needed to support the payload.
Leading edge	Smooth, suitable radius	Can help provide predictable behavior as the angle of attack increases.
Trailing edge	Clean and well-shaped	Helps manage drag and airflow separation.
Overall wing design	Adequate area and suitable aspect ratio	A larger wing area can support more weight at a given speed; aspect ratio affects induced drag.

AIRFOILS GEOMETRY EFFECT ON THE CASE (B) AIRCRAFT:

☐ **Turns:** Depend on lift capability, wing loading, bank angle, and available control.

☐ **Rolls:** Depend strongly on aileron size, placement, and control authority, along with wing inertia.

☐ **Loops**: Depend on pitch control, speed, lift, available thrust, and structural strength.

Geometric feature	Preferred characteristic	Reason
Camber	Moderate or relatively low, depending on the design	Can provide a useful balance of lift, drag, and pitching moment.
Thickness	Sufficient for structural strength without unnecessary weight	Helps the wing withstand maneuvering loads.
Symmetry	A symmetric profile may be advantageous for aerobatics	Provides more similar behavior at positive and negative angles of attack.
Leading edge	Appropriate for the expected angle-of-attack range	Influences stall behavior and control predictability.
Trailing edge	Compatible with effective ailerons and other controls	Ailerons are essential for controlling roll; the elevator controls pitch.

Discuss whether one airfoil can satisfy both missions, including the trade-offs.

Yes, one airfoil can serve both missions to a reasonable extent, but it will involve trade-offs. Let us see how different parameters act in both the cases and what are its trade-offs:

Parameter	Maximum payload and stability	Maneuverability	Trade-off
Camber	Moderate to high camber can help generate lift at low speeds	Moderate or lower camber may be useful for balanced positive and negative flight	More camber can improve low-speed lift but may increase pitching moment and drag in some conditions
Thickness	Sufficient thickness for a strong wing structure	Enough thickness to maintain strength without excessive structural weight or drag	A stronger wing may add weight if not designed efficiently
Lift coefficient (C_L)	High lift helps support the aircraft and payload	Adequate lift is needed for tight turns and loops	High maximum lift alone does not guarantee fast or responsive maneuvers
Drag coefficient (C_D)	Lower drag improves endurance and reduces power demand	Lower drag helps retain speed during maneuvers	The best efficiency may occur at a different operating point from maximum lift
Stall behavior	Predictable stall behavior supports safe, controlled flight	Predictable behavior at high angles of attack supports control during demanding maneuvers	Stall characteristics must be assessed over the aircraft's operating range
Pitching moment (C_m)	Manageable pitching moment helps with trim and stability	Manageable pitching moment supports pitch-control authority	The tail and center of gravity must be designed around the airfoil

I would use **NACA 2412** as the best compromise candidate amongst the total of six candidates.

- Moderate camber can provide useful lift during normal flight.
- Its 12% thickness provides a practical starting point for wing structural design.
- Your XFLR5 plots suggest favorable peak lift-to-drag efficiency among the three candidates.
- It is worth testing for both payload-carrying flight and general maneuverability.

We can reduce the trade-offs by keeping the basic airfoil the same but changing some other features like:

• **Wing area**: Choose enough area to support the intended payload at the target flight speed.

• **Aspect ratio**: Balance induced drag against the roll response and structural requirements.

• **Control surfaces**: Use appropriately designed ailerons for rolls and an elevator for pitch control.

• **Center of gravity and tail**: Design these for suitable stability and control authority.

- **Structural design:** Make the wing strong enough for the intended maneuvering loads without adding unnecessary mass.

Distinguishing Airfoil Characteristics from Whole-Aircraft Factors

Comparison between Airfoil and Aircraft Characteristics and table containing some important factors affecting the performance of the aircraft:

A. Airfoil characteristic: These describe the cross-sectional shape of the wing and its aerodynamic behavior.

- **Camber**: Influences lift and pitching moment.
- **Thickness**: Influences aerodynamic behavior and provides space for wing structure.
- **Lift coefficient**: Indicates how effectively the airfoil generates lift under specified conditions.
- **Drag coefficient**: Indicates its aerodynamic resistance.
- **Stall behavior**: Describes how lift changes when the angle of attack becomes too high.
- **Pitching moment**: Describes the airfoil's tendency to rotate nose-up or nose-down.

B. Whole-aircraft factors: These describe the design and behavior of the entire RC plane.

- **Wing loading**: Aircraft weight divided by wing area.
- **Wing geometry:** Wingspan, aspect ratio, taper, and twist.
- **Control surfaces**: Ailerons, elevator, and rudder.
- **Structure**: Wing strength, stiffness, and total mass.
- **Thrust and power**: Motor, propeller, battery, and available thrust.
- **Stability and balance**: Center of gravity, tail size, and tail arrangement.

Factor	What it affects	Example in an RC plane
Airfoil camber	Lift at a given angle of attack	A cambered wing may generate useful lift at a relatively low speed.
Airfoil thickness	Aerodynamic behavior and structural packaging	A thicker section may provide more room for a spar.
C_L and C_D	Lift and aerodynamic resistance	Higher lift can help support weight; lower drag can reduce power demand.
Wing loading	Stall speed and turning performance	A heavily loaded wing generally requires more speed to produce sufficient lift.
Aspect ratio	Induced drag and wing geometry	A higher aspect ratio can reduce induced drag, but may affect roll response and structural design.
Ailerons and elevator	Roll and pitch control	Ailerons control rolling; the elevator controls pitching.
Structure and mass	Strength, stiffness, inertia, and load capacity	A strong but unnecessarily heavy wing can reduce payload capacity and agility.
Thrust and power	Acceleration, climb, and energy available during maneuvers	More available thrust can help the plane regain speed after a maneuver, within its design limits.
Center of gravity and tail	Stability and control	Incorrect balance can make the plane difficult to control regardless of airfoil choice.

NOTE: AIRFOILS ARE AN IMPORTANT DETERMINANT OF THE PERFORMANCE OF AIRCRAFT BUT THEY ARE NOT THE SOLE DETERMINANTS THERE EXIST VARIOUS OTHER FACTORS THAT ARE ALSO A SIGNIFICANT FACTORS IN CONTRIBUTING TOWARDS THE PERFORMANCE OF AN AIRCRAFT.